

UNIT 3

Pollution and Pollution Control

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Environmental Crises

The laws of energy give keys to understanding the environmental crises currently facing the world.

Environmental Crises:

- Global warming
- Air pollution
- Indiscriminate waste disposal
- Land degradation
- Resource depletion
- Climate change
- Soil pollution
- Famine
- Flooding
- Polluted water supply
- Ozone layer depletion
- Destruction of rainforest
- Desertification
- Rapid population growth
- Disease pandemic
- Water scarcity

Pollution

Pollution is defined as an **undesirable change in the chemical, physical and biological characteristics of air, water and land** which tends to harmfully affect life, health, survival, or activities of humans or other living organisms.

- Under this definition, pollution does not necessarily have to cause physical harm. It may merely interfere with human activities, e.g. a lake may be considered polluted if it cannot be used for boating activities.
- The term 'undesirable change', requires value judgements.
- An alteration may be judged favourable by some; or the undesirable effect may be considered acceptable when compared with the favourable effect e.g. an affluent country may ban the use of DDT as a pesticide because of a judgement that the risks (especially to non-human organisms) outweigh the benefits.
- At the same time, a country with insufficient food production or a country where malaria affects much of the population may decide that the advantages of using DDT to kill crop pests or malaria-carrying mosquitoes outweigh the risks of its undesirable effects.

Pollutants

Pollutant -Any substance that causes pollution

Classified based upon the pollutants susceptibility to degradation:

- Degradable pollutants
- Non-degradable pollutants
- **Degradable Pollutants**: These are the pollutants that can be decomposed, removed, consumed and thus reduced to acceptable levels either by natural processes or by human-engineered systems.
- There are 2 types of degradable pollutants:
 - Rapidly degradable and slowly degradable pollutants

Pollutants

- **Rapidly degradable pollutants** – decompose faster e.g. Human sewage, animal and crop wastes.
- **Slowly degradable pollutants** – decompose slowly but eventually are either broken down completely or reduced to harmless levels, e.g. DDT, radioactive materials (strontium-90, plutonium-239 etc.).
- **Non-degradable pollutants**: pollutants which are not broken down by natural processes. E.g. heavy metals (like lead, mercury) and some plastics etc.

Classification of Pollutants

One classification of pollutants put pollutants into three divisions.

- The first type of pollutants include those substances that **occur in nature**, but as a result of human activity are found in **unusually large concentrations** e.g. CO₂.
- The second type of pollutants are the **toxic compounds** that are released into the environment as a result of **anthropogenic activities**. These compounds are **not found naturally** in the environment e.g. mercury, arsenic, selenium etc.
- The third type of pollutants occur when substances which are **not themselves toxic** but are released into the environment as a **result of human activity**. E.g. heat from power plants and industries, Salt, colours, Fe.

Pollution

There are 3 main types of pollution

- Air pollution
- Water pollution
- Soil/Land pollution

Others are:

- Noise pollution
- Ocean pollution

Air Pollution

Air pollution is air contaminated with one or more substances at concentrations harmful to humans, other animals, vegetation or materials.

There are 2 types of air pollutants

- **Primary Air Pollutant** -Normally defined as a chemical or substance that is **added directly to air** at harmful concentrations eg. CO₂ above its normal concentration or release of lead from vehicles, suspended particles, CO, NO₂.
- **Secondary Air Pollutant**-Harmful chemical or substance **formed** in the atmosphere through a chemical reaction among air components. E.g. O₃, SO₃, H₂O₂, HNO₃

Sources:

- Human activities. Eg. Transportation, power and heat generation, industrial processes and the burning of solid wastes
- Naturally occurring events. Eg. Volcanic eruptions, fires from lightning and storms

Major Air Pollutants

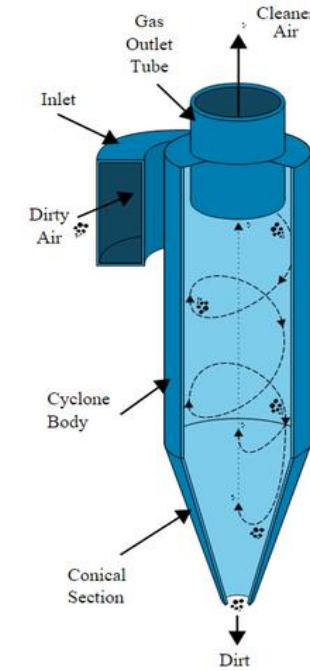
- **Oxides** of carbon, sulphur and nitrogen: carbon monoxide (CO), Carbon dioxide (CO₂), sulphur dioxide (SO₂), sulphur trioxide (SO₃), nitrous oxide (N₂O), nitric oxide (NO), nitrogen dioxide (NO₂)
- **Hydrocarbons** and volatile organic compounds: methane, benzene, solvents, gasoline
- **Photochemical oxidants**: ozone, aldehydes
- **Particulates**: smoke, dust, oil, asbestos, Pb, Hg, Be, Smog [industrial (gray) & Photochemical (brown)]
- **Radioactive** substances: radon, emission from electric power plants and fossil fuel
- Other **inorganic/ organic compounds**: hydrogen sulphide, nitric acid, pesticides, herbicides, alcohols.
- **Biological** substances: pollen, mould

Effects of Air Pollution

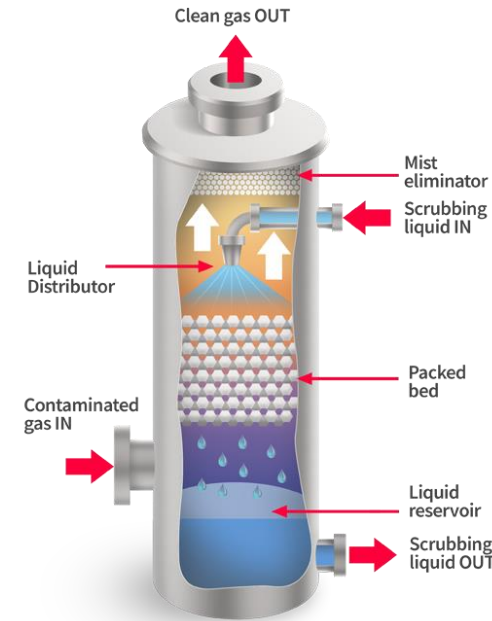
- Affects **human health** - bronchitis, emphysema, tuberculosis and lung cancer
- Causes unpleasant fumes and dust particles **reducing visibility**
- Damage to **crops** and vegetation
- Causes **acid rain** which can increase the acidity of some lakes- harmful to aquatic life
- Air pollutants such as CO, CH₄, CO₂ causes **global warming and destruction of the O₃ layer**.
- CO impairs ability to **learn, visual perception and manual dexterity**
- Causes damage to property due to **corrosive** nature of some air pollutants. Eg. SO_x, NO_x

Control Devices for Air Pollution

- **Wet scrubbers** –can remove 99.5% of particulates and 80 to 95% SO_2 by passing exhaust through liquid such as water
- **Cyclone separators** – removes 50 – 90% of large particles by swirling exhaust gases through a funnel shaped chamber which collects the particles through centrifugal forces.
- **Baghouse filters**: removes about 99.9% particles from exhaust gases when passed through fibre bags.



Cyclone separators



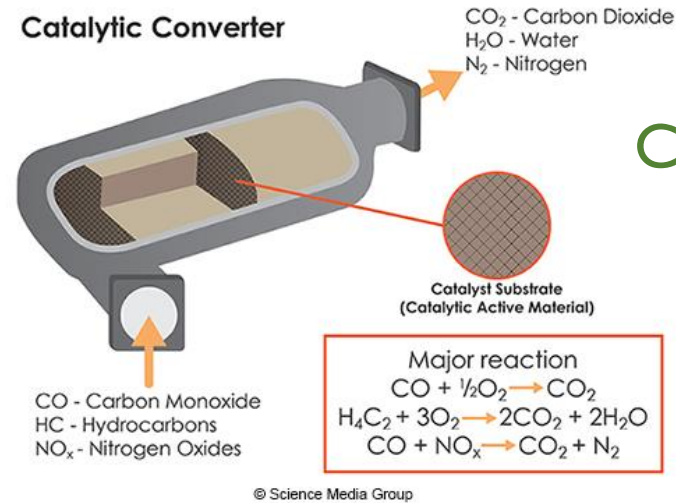
Wet scrubbers



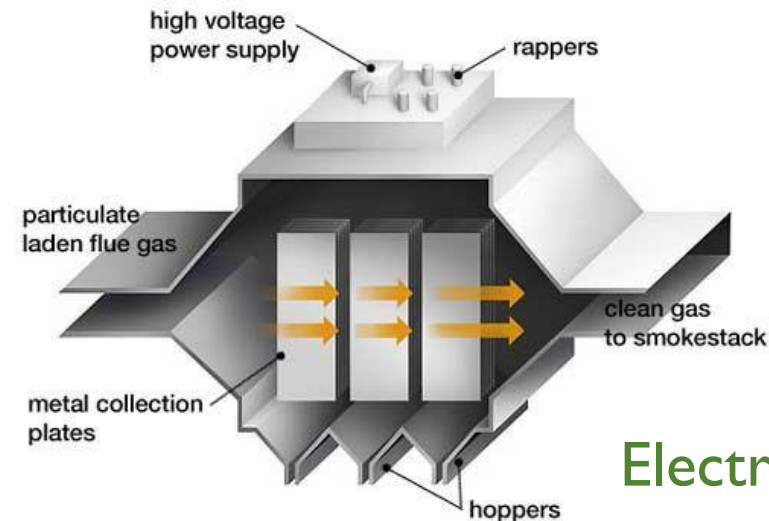
Baghouse filters

Control Devices for Air Pollution

- **Stack heights electrostatic precipitators**
 - removes 99.5 % of large particles through an electrostatic force that charges the particles so that they can be attracted to a series of electrodes and be removed.
- **Natural washout by rain**
- **Catalytic converters**



Catalytic converters



Electrostatic precipitators

Other Control Measures - Air Pollution

- Increase ventilation
- Decrease use of vehicles.
- Use of unleaded compounds eg. Unleaded gasoline
- Decrease use of coal
- Limit SO₂ and NO₂ emissions
- Use formaldehyde free materials
- Afforestation
- Prevent vaporization during fueling.

Global Warming

- Light absorbed by the earth is converted to heat.
- CO₂ and other gases (NO₂, O₃ and CH₄) in the troposphere trap some of the heat and warms the earth
- This allows the earth to maintain a warm temperature and support life
- Greenhouse effect warms the atmosphere about 38°C on average.
- If there is no green house effect, the earth will be much colder.
- Enhanced global warming is the gradual increase of CO₂ concentrations in the atmosphere.
- Human activities (particularly from industries, deforestation, burning of fossil fuels) is causing increase in CO₂ – the earth is becoming too warm.

Effects of Global Warming

- High temperatures on the earth
- Changes to weather patterns and regional climates
- Natural important agricultural areas could become less productive
- Natural ecosystems would also be affected.
- Leads to rising sea levels – can cause flooding, erosion and increase in salinity of groundwater bodies and estuaries.
- Diseases Eg. In 1878, atmospheric warming caused malaria epidemics in North America. 100,000 cases of malaria were reported and 28,000 people died.

Remedying Green House Effects

- Taxing of fuels (which uses carbon) to generate money for the government
- Efficient utilisation of fossil fuels
- Utilise renewable energy systems
- Afforestation

Ozone

- Ozone is a colourless gas with a pungent odour, and the chief component of smog.
- Ozone could exist in the troposphere or the stratosphere.
- Ozone is produced in the atmosphere (troposphere) when sunlight triggers certain chemical reactions.
- The precursors, or chemicals that are initially needed for this reaction to take place include volatile organic compounds (VOCs) and nitrogen oxides (NO_x).
- VOCs are released into the air when petroleum products are combusted, handled or processed. NO_x are also produced by combustion. Sunlight provide energy to fuel the reactions between the VOCs and the NO_x and naturally occurring atmospheric gases, resulting in the production of ozone and eventual generation of smog

Ozone in Stratosphere

- The ozone layer in the stratosphere filters out some of the UV light from the sun.
- Destruction of the ozone layer allows more UV rays to reach us.
- Several greenhouse gases (e.g. Chlorofluoro-carbons(CFCs) and N_2O) are ozone depleting gases.
- Tropospheric warming apparently induces stratospheric cooling that exacerbates ozone destruction. As more of Earth's radiant heat is trapped in the lower atmosphere, the stratosphere cools further, enhancing the catalytic destruction of ozone. Further, the loss of ozone itself augments the cooling of the stratosphere.
- Interactions between climate change and stratospheric ozone may delay recovery of the ozone layer by 15 – 20 years.

Depletion of the ozone layer will create the following problems:

- Human skin cancer, cataracts
- Reduction in populations of certain fish larvae and plankton.
- It also reduces the useful life of outdoor paints and plastics.
- Damage to plant and animal life including food crops
- Alteration in global climate patterns

Acid Rain

Causes

- Emissions of sulphur dioxide (SO_2) and nitrogen oxides (NO_2) (mostly from motor vehicles and coal burning power plants) interact with sunlight and water vapour in the upper atmosphere to form acidic compounds (sulfuric acid and nitric acid).
- These compounds are removed from the air often through rainfall and so the name acid rain or acid snow
- The compounds can also join dust or other dry airborne particles and fall as dry deposition (acid dust).

Acid Rain

Effects:

- Acid rain increases acidity of surface waters so much that it reduces or eliminates their ability to sustain aquatic life.
- Forest and agricultural crops are vulnerable because acid deposition can leach nutrients from the ground
- Can hamper the activities of micro-organisms that nourish plants (rhizobium, etc)
- Release toxic metals that would normally be tied up in the soil at higher pH.
- The pollutants that cause acid rain can also make the air hazy or foggy affecting vision, communication and navigation.

Acid Rain

Control:

- Power plants burning coal have to reduce release of SO_2 . Total SO_2 released should be permanently limited to the low levels set.
- Stiff penalties for industries that release more pollution
- Bonus allowance to be given to power plants that install clean coal technology designed to reduce SO_2 release or for using renewable energy sources such as solar or wind.
- New cars and utility boilers that release lesser NO_x emissions to be designed.

Destruction of
vegetation due to acid
rain in North Carolina,
USA



Soil / Land pollution

Soil pollution is the occurrence of a chemical or substance in a soil at levels detrimental to living organisms or at levels above its permissible threshold in soils.

Sources:

- Leachates from landfills
- Acid rain
- Volcanic eruptions
- Leakage of oil from oil refineries.
- Excessive application of herbicides, pesticides, fertilizers to agricultural fields
- Discharge of untreated industrial (eg. Mining) or domestic solid and liquid waste
- Rupture of underground storage tanks
- Irrigation with saline water, wastewater
- Salt intrusion and dissolved chemicals from the sea or surface water
- Weathering of rocks – Zn, Rn, As
- Burying of waste eg. Plastics, hazardous materials.

Common Pollutants

- Heavy metals: Pb, Cd, Hg, As
- Petroleum hydrocarbons
- Pesticides, fertilizers and herbicides
- Solvents
- Pharmaceuticals
- Plastics
- Livestock residues containing heavy metals, veterinary antibiotics, pathogens
- Food wastes



Effects of Soil Pollution

- Increased erodibility of soil
- Loss of soil and its nutrients, low crop yield
- Destruction of soil habitat
- Loss of biodiversity in soil
- Seepage of pollutants into water bodies, drinking water sources
- Increased soil salinity and toxicity
- Clogged drains
- Flooding
- Diseases – malaria, cholera



Control of Soil Pollution

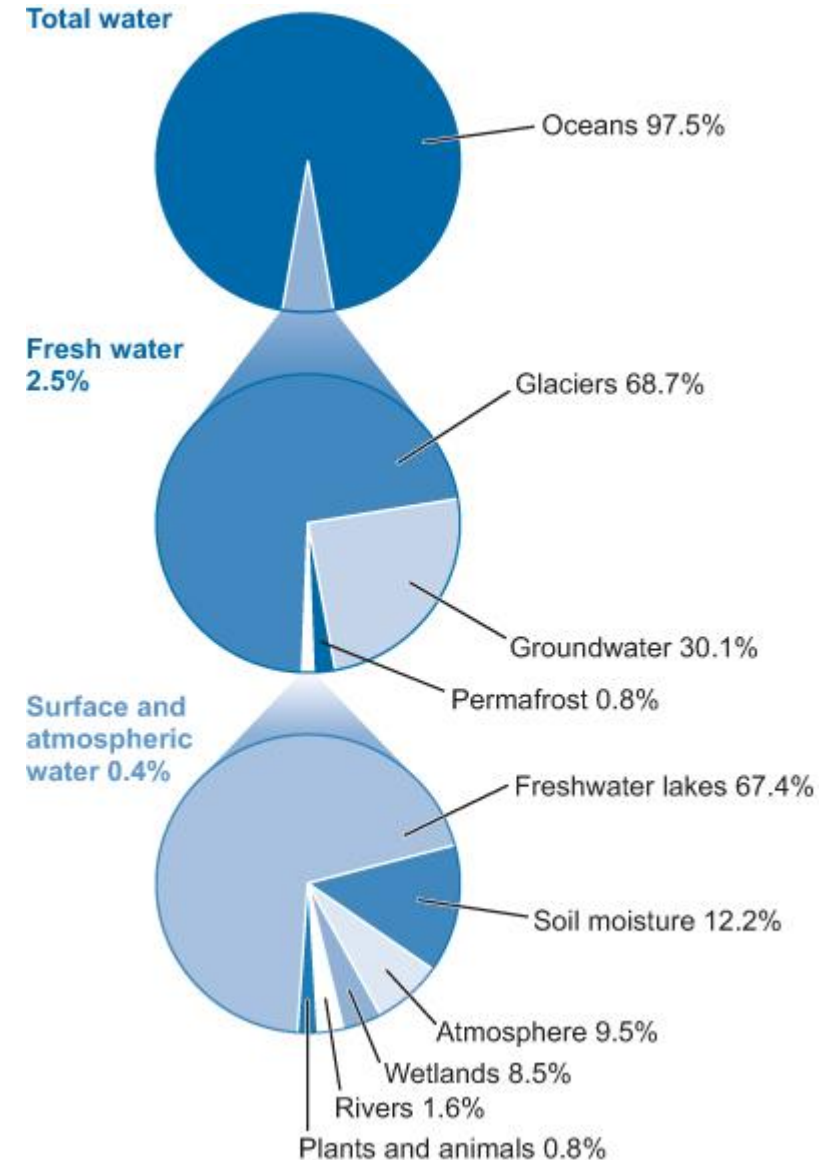
- Treatment of domestic/industrial wastewater prior to discharge into the environment
- Control of oil spills at oil refineries and taking prompt actions if they occur
- Researching into oil and plastic eating bacteria
- Reduction in chemical fertilizer, pesticides and herbicide use on farming lands
- **Reduce, reuse, recycle (3Rs)** materials such as glasses, plastics rather than disposing them off. Eg. Recovering 1000kg of papers can save 17 trees.
- Afforestation
- Provide regulations on waste discharges
- Construction of engineered landfills
- Prevention bureau
- Prevention/reduction in the use of phosphorus in detergents

Water Distribution on Earth

- About 71% of earth is covered with water
- Total amount of water on earth is 1386million Km³
 - Saline water= 97.5%
 - Fresh water=2.5%

The freshwater is made up of:

- Ice and permanent snow cover=68.7%
- Fresh underground water= 30.1%
- Permafrost = 0.8%
- Surface and atmospheric water = 0.4%



Uses of Water

- Domestic use
- Agricultural use
- Industrial use
- Recreation
- Power production (hydropower)
- Transportation (navigation)

Water Pollution

- Water pollution is a harmful modification of water caused by the addition of substances likely to modify its quality, aesthetic aspect and use for human purposes. The polluting agent may be physical, chemical or biological in nature and may cause discomfort, nuisance or contamination
- Water has very unusual characteristics, which make it very ideal for the survival of living things
- Dissolves many substances more than any solvent in the world hence it is easily polluted

Major Water Pollutants:

- Pathogens – bacteria, viruses, fungi, parasitic worms, sewage (waste water from toilets, bathrooms and kitchen)
- Plant nutrients- N, PO_4^- from fertilisers, manure
- Off shore drilling produce water
- Radioactive wastes
- Pesticides eg. DDT
- Cleaning solvents, detergents
- Heavy metals
- Oil, gasoline
- Solid waste
- Acids, salts
- Sediments



Effects of water pollution

- Nuisance and destruction of aesthetic value
- Eutrophication: Over growth of aquatic plants from plant nutrient (nitrogen and phosphorus) as a result of pollution
- Damage to plant and animal life
- Human genetic and reproductive damage caused by radioactive, pesticides and industrial chemicals
- Property damage caused by muddy and corrosive waters

Contaminated stormwater heavily polluting rivers



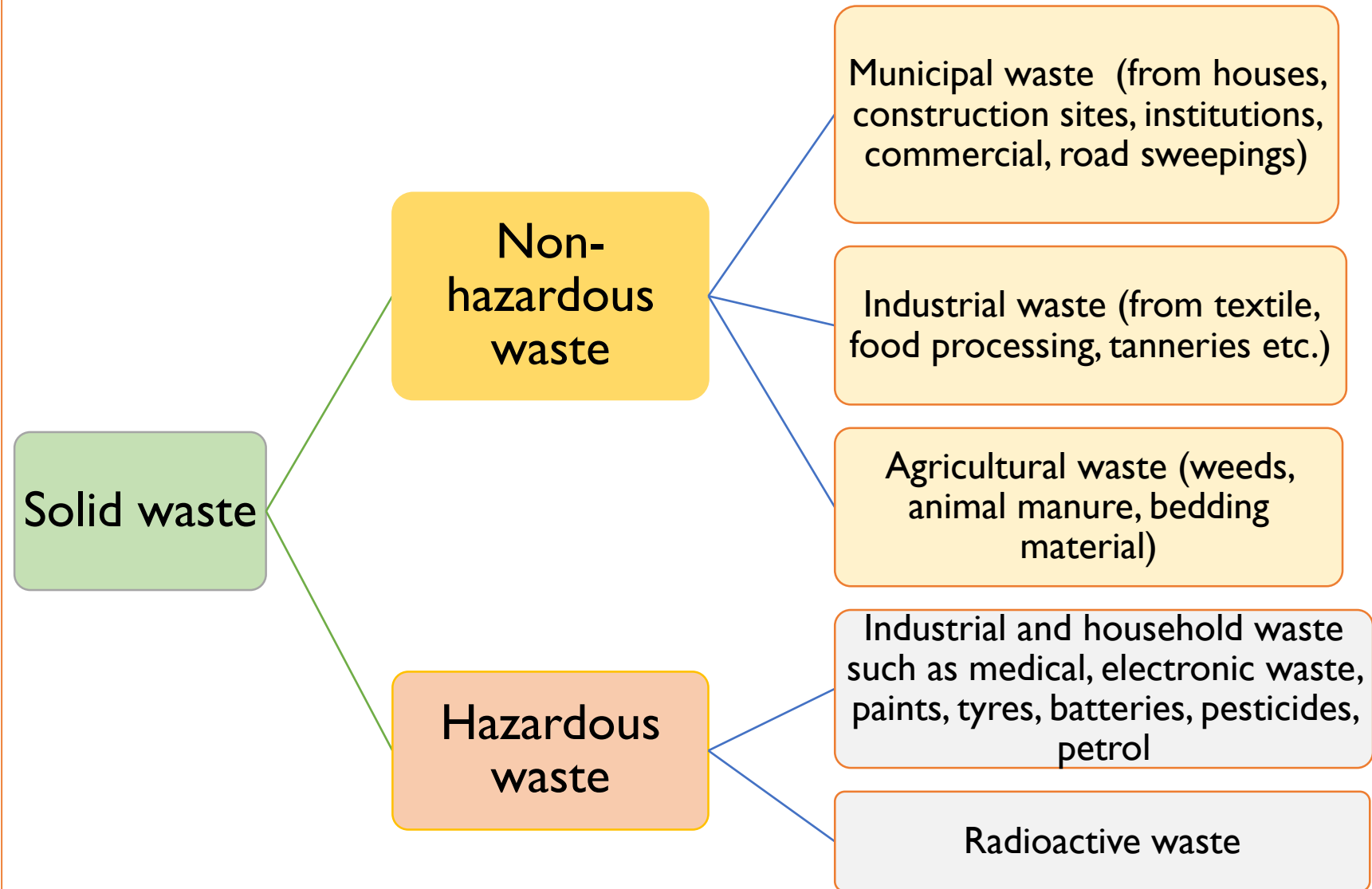
Control of water pollution

- ❖ Construction of wastewater treatment
- ❖ Treating industrial and domestic waste prior to discharge into stream or water bodies.
- Recycling and reuse of water
- Enacting and enforcing National laws to reduce pollution (EPA guidelines)
- Reducing phosphates content in detergent
- Oil spills controlled by chemical foams and absorption pads
- Research into the finding of oil eating bacteria

Solid waste

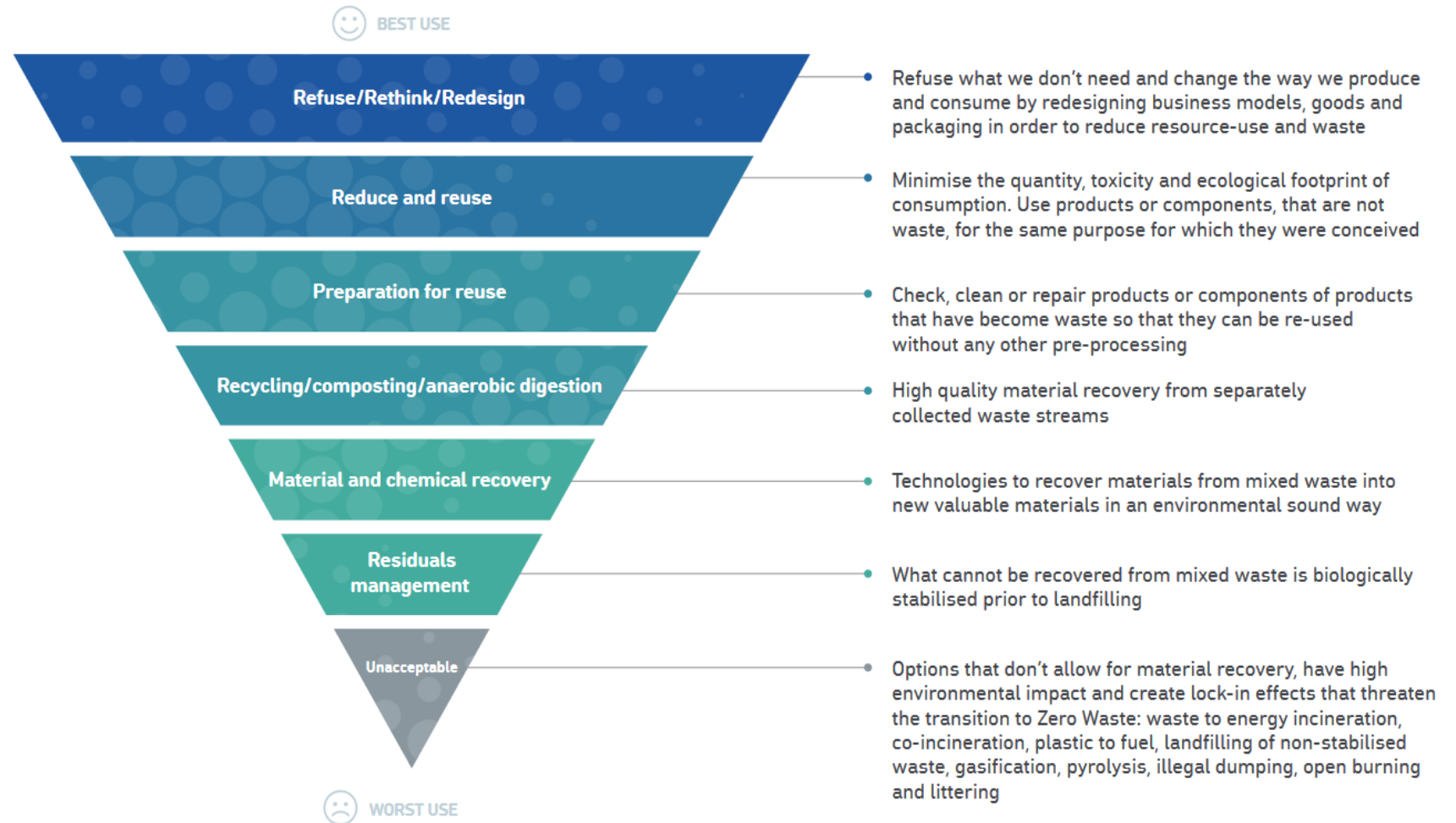
❖ Solid waste is a useless, unwanted or discarded material which is not liquid or gas.

❖ Solid waste can be grouped under hazardous and non-hazardous waste.



Management of Solid Waste

- ❖ Solid waste management involves collection, transport, treatment and disposal of solid waste.
- ❖ Zero waste concept: focuses on waste prevention.
- ❖ Aims at reuse of all products such that no waste is generated for disposal on land, air or in water



Zero Waste Management Hierarchy

Management of Solid Waste

- ❖ **Refuse/Rethink/Redesign**: involves any strategy that stops the production of waste such as use of less raw material and hazardous material in manufacture, rethinking design and production models to make them waste-free. Also making more products durable.
- ❖ **Reduce and Reuse**: continual use of a used product so it does not become a waste product
- ❖ **Preparation for reuse**: Waste materials are examined, cleaned, repaired, refurbished so they can be used again.
- ❖ **Recycling/ composting/anaerobic digestion**: turning a waste product into a new material. Waste materials are collected separately eg. Recycling plastic waste
- ❖ **Material and Chemical Recovery**: extraction of minerals or energy from a waste product.
- ❖ **Residuals management**: Biological stabilization of waste to reduce mass and volume of waste taken to landfills to minimise landfill gases and leachates.
- ❖ **Disposal**: disposal of waste via landfilling of unstabilised waste or incineration

Problems of Solid Waste Management in Ghana

- a. Lack of political will to enforce sanitation by laws
- b. Facilities for solid waste collection not available
- c. Lack of solid waste disposal sites
- d. Lack of willingness by some communities to pay for the management services
- e. Cannot afford to pay for services
- f. Government by law is responsible for waste management but commitment has not been to good
- g. Lack of awareness of the citizenry on the health impact of poor solid waste management

Wastewater

- It is spent water after domestic, institutional, commercial or industrial, agricultural use of water. It also includes storm water or surface runoff.
- The three types of wastewater are:
 - **Domestic** wastewater: wastewater from homes, institutions specifically from kitchens, bathrooms and laundries
 - **Industrial** wastewater: wastewater from industries eg. Tanneries, beverage, pharmaceuticals, textile factories
 - **Storm** water: Surface runoff produced from heavy rains or snowmelts.



Types of Domestic Wastewater

- **Blackwater**: Wastewater from toilets. Contains urine, faeces, flush water and cleansing materials.
- It therefore, contains lots of pathogens (disease-causing microorganisms) and nutrients.
- Blackwater requires treatment before it can be reused.
- **Greywater**: wastewater from sinks, washing machines, bathrooms, bathtubs. It is less contaminated and can be reused.

Characteristics of wastewater

Three main characteristics of wastewater

Physical

- Temperature
- Colour
- Odour
- Turbidity
- Total solids

Chemical

- pH
- Chemical Oxygen Demand
- Biochemical oxygen demand
- N, P, Cl-,
- Heavy metals (Zn, Pb)
- Oil, grease

Biological

- Microorganisms like:
- Helminth eggs
- Bacteria (E-coli)
- Viruses
- Protozoa

Treatment of Wastewater

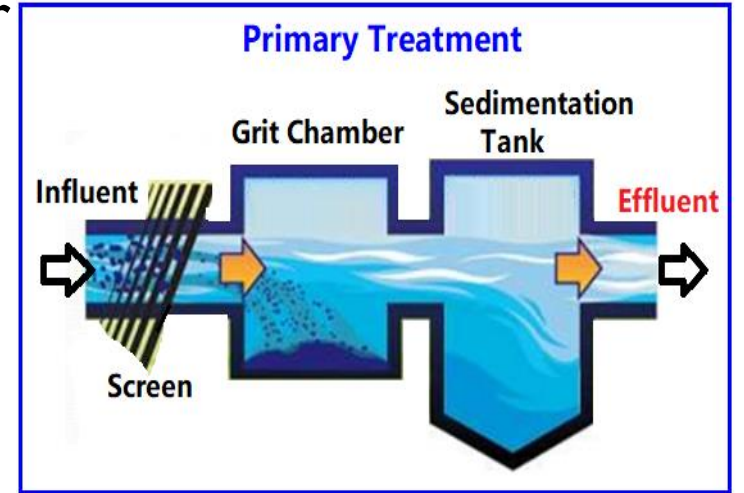
Primary treatment: to remove materials which either floats or settles easily under gravity.

Larger materials like rags, sticks are separated from the wastewater used bar **screens**.

Grit chamber is meant to take out sand, silt, stones or egg shells.

Primary sedimentation tank takes out the particles which readily settles.

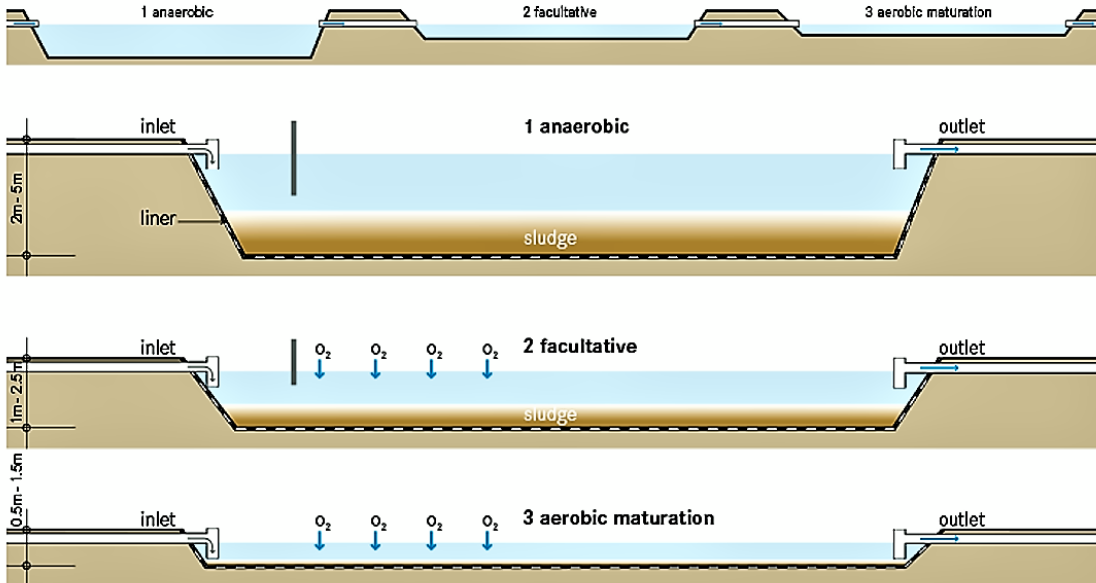
The wastewater then goes through a secondary treatment



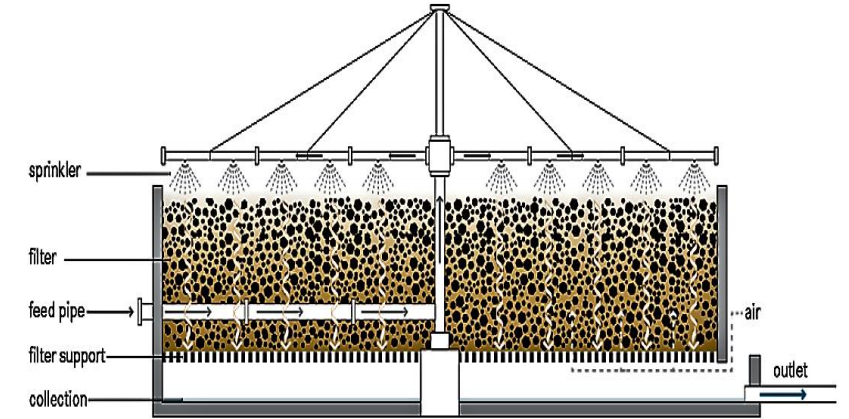
Treatment of Wastewater

Secondary treatment: to remove biodegradable matter from the wastewater. Microorganisms breakdown the organic matter in the wastewater.

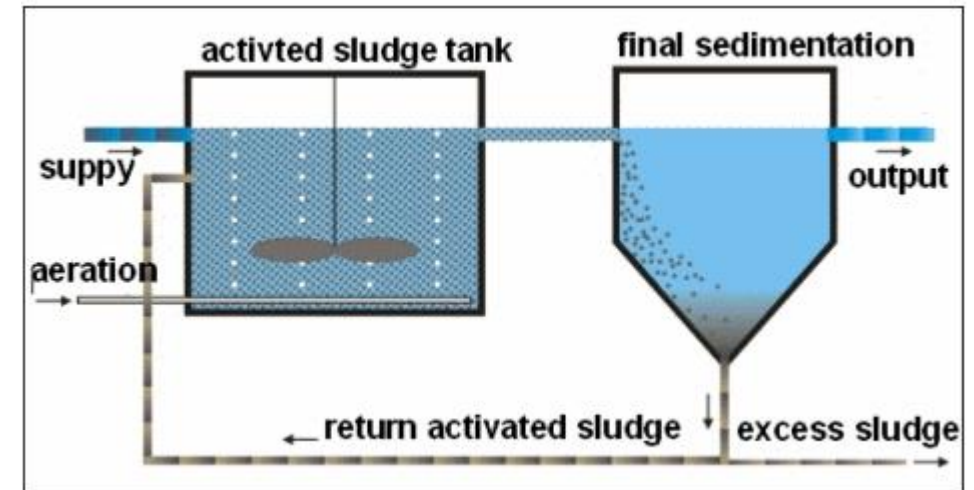
Wastewater stabilisation ponds



Trickling filter



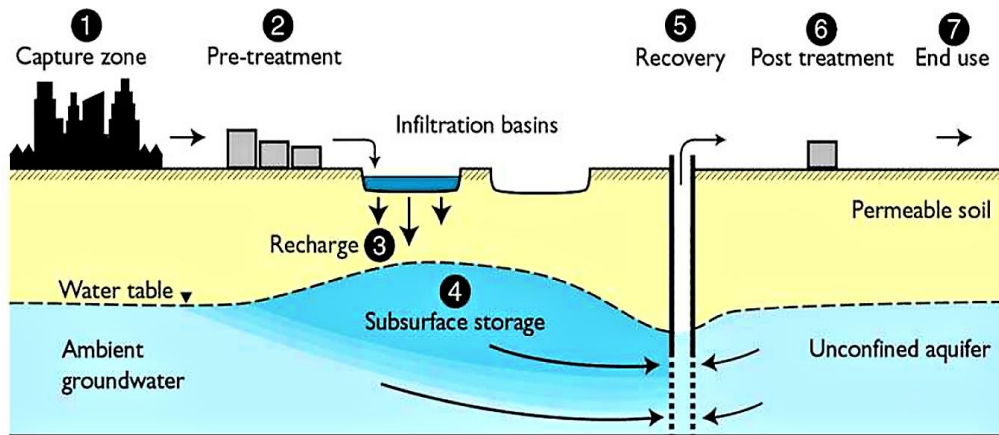
Activated sludge



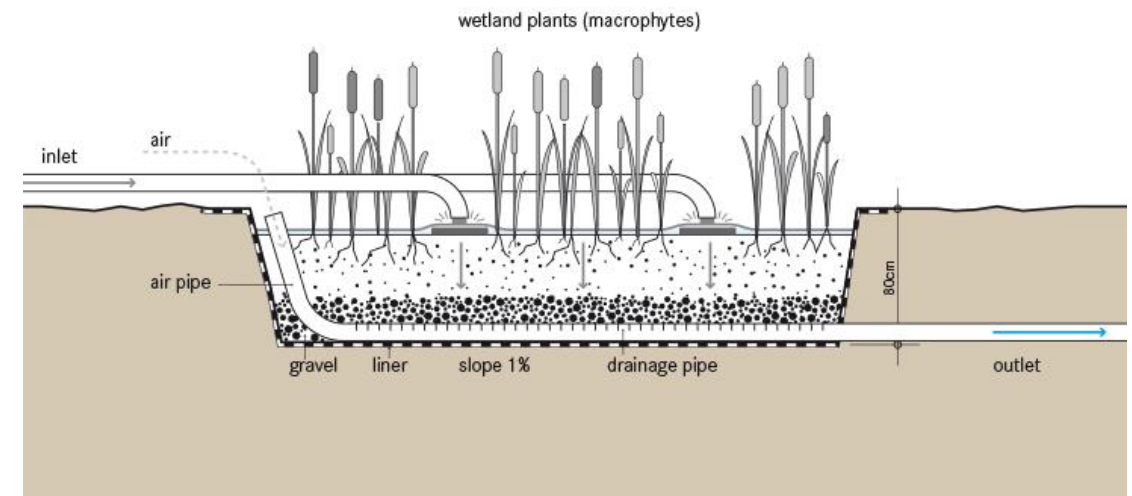
Treatment of Wastewater

Secondary treatment: to remove biodegradable matter from the wastewater. Microorganisms breakdown the organic matter in the wastewater.

Soil aquifer treatment



Constructed wetlands



Ocean Pollution

- ❖ Oceans are the major source of water vapour which returns to land as precipitation.
- ❖ About 10% of the protein taken by humans is provided by the oceans.
- ❖ One third of the global oxygen supply is produced from oceans.

Ocean pollution results from:

- Use of oceans as dumping grounds for wastes (pesticides, heavy metals, sewage sludge, radioactive waste, plastics, construction waste etc.)
- Polluting it with oil (by oil tankers and during oil drilling exploration)
- Over exploitation of its marine resources

Noise pollution

Noise pollution, unwanted or excessive sound that can have deleterious effects on human health and environmental quality.

Sources

- Traffic including air traffic eg. highway, railway, and airplane traffic
- Animals eg. Barking dogs
- Bars, restaurants, discotheques, pubs, clubs, Churches and funeral grounds
- Household chores and gadgets: vacuum cleaners. Washing machines, blenders
- Construction activities

Our studies show that the sound level in many of the churches around KNUST and environs are above 85 decibels the maximum permissible level

Effects of Noise pollution

Excessive noise can cause:

- Annoyance
- Trouble communicating
- Nervous breakdown
- Physical or mental deterioration, bad memory, inability to concentrate
- Sudden noise can cause psychic shock, constriction of blood vessels, pupil dilation, tensed muscles, increased heartbeat, wincing and stomach spasms.
- Disruption of activity
- Partial or total loss of hearing
- Sleeping disorders

Control of Noise pollution



Ear protectors



Ear plugs

- Reduce noise at its source
- Replacing noisy machines and operations with less noisy ones.
- Reducing the amount of noise entering the listener's ears by wearing ear plugs or ear protectors.
- Production of quieter trucks, motorcycles and other machines.
- Enactment and enforcements of tougher legislations.
- Factories and transport terminals should be sited far away from residential areas.
- Turning off or lowering the volume of appliances
- Planting trees along roads to absorb noise from vehicles.
- Cars can have separate city and highway horns. Drivers should limit the amount of honking.

Personal Hygiene

- Personal hygiene is a measure taken by an individual to promote cleanliness to prevent transmission of diseases from source to susceptible host.
- Includes: body, teeth, clothes, hands, ears, foot etc.
- Skin care
 - Bathing the whole body including the hair well every day or after sweating or getting dirty.
 - Special attention to ears, armpit, genitals, anal areas.
 - Dry the body thoroughly with clean towel
 - Change into clean clothes after bathing
 - Avoid sharing soaps, sponges and towels.

Personal Hygiene

- **Oral hygiene:** the mouth provides good environment for bacterial growth
 - Rinse mouth after each eating
 - Brush teeth with fluoride toothpaste twice daily- before breakfast and before sleeping in the evening.
 - Floss your teeth.
- **Face hygiene:**
 - Wash face every morning.
 - Special attention given to eyes



Dental Floss

Personal Hygiene

- **Hand care:** handwashing plays a significant role in preventing faecal-oral disease transmission.
- Keep fingernails short and clean.
- Handwashing must be done with soap/ash under running water
- Should be done before handling food, after eating, touching nose or contaminated places, visiting the toilet, touching raw food (meat).



Personal Hygiene

- Foot hygiene:

- The feet should be carefully washed and dried at least twice daily.
- Toenails should be kept short.
- Put on footwear as often as possible.
- Socks and shoes (if possible) should be washed and sun-dried.

Cloth hygiene:

- Change dirty clothes for new ones each day.
- Dirty clothes should be washed with adequate clean water and soap and ironed if possible.

Personal Hygiene

- Genitalia
 - It is more hygienic and safer not to shave off pubic hairs. But can keep it low.
 - Use low-scented and mild soap.
- Menstrual hygiene:
 - Wash the outer genitalia with clean water daily.
 - Change sanitary pads or tampons regularly.
 - Wash hands before and after handling pads/tampons.
 - Used pads should be buried or disposed of appropriately.
 - Dry undergarments in the sun.

Domestic Hygiene

- Domestic hygiene – cleanliness in the home.
- Ensure good ventilation
- Sweep and clean floors, weed around houses
- Wash dishes after meals
- Store food and water hygienically
- Dispose off solid and liquid waste well.
- Good drainage of wastewater
- Use of improved toilets
- Use insecticide to keep away pests/ Sleeping in treated nets

Food Hygiene

- Wash hands well before and after handling food.
- Wash ingredients (fruits and vegetables) before cooking.
- To prevent cross-contamination, raw ingredients should be separated from cooked ones, utensils, chopping boards, cutlery etc used for raw food (meat, eggs) should be separated from those used for cooked food.
- Wash and sanitize all equipment, cooking pans and surfaces before food preparation. Keep them clean and dry after use.
- Wear protective clothing, keep hair covered, do not sneeze, spit, touch face or chew gum during cooking.

Sanitize surfaces



Food Hygiene

- Thaw food inside the fridge, under cold water, in a microwave or at room temperature.
- Lids of canned food should be cleaned before opening.
- Note expiry dates on packaged food before use.
- Check packaging for appropriate food storage.
- Have a cleaning plan for the kitchen.
- Mouldy foods should be discarded immediately.
- Mop kitchen floors when wet and after food preparation.
- Grains, dried and powdered foods should be kept away from liquids and from one another.



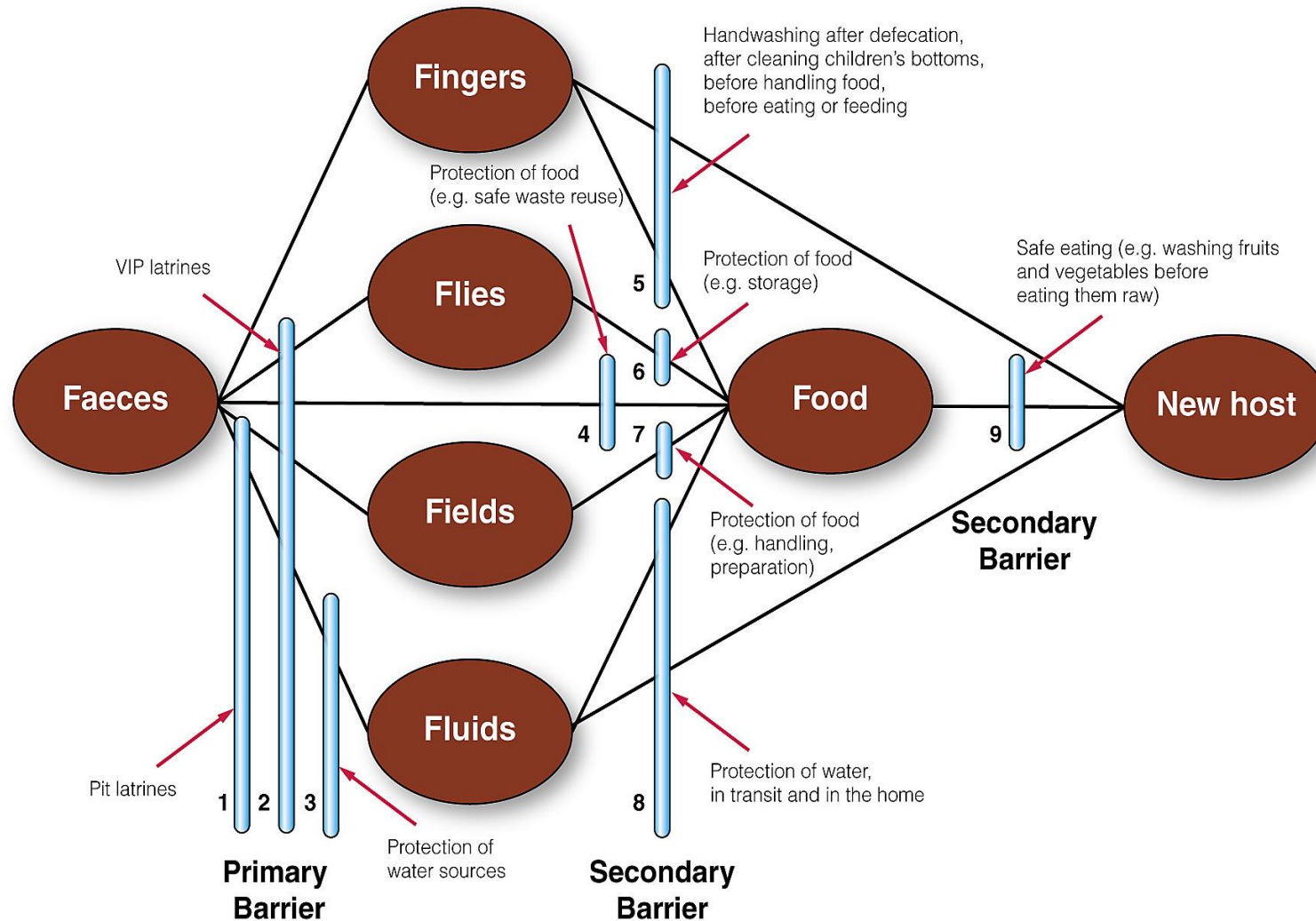
Mouldy groundnuts

Food Hygiene

- Good quality water (potable) should be used for food preparation
- Cook food thoroughly to kill pathogens before eating
- Put food in refrigerator right away or at most 2 hours after preparation.
- Refrigerate food/ ingredients at the right temperature to prevent bacteria growth
- Cover food or keep them refrigerated.
- Kitchen waste should be kept far from the cooking area. Empty and clean trash cans or dustbins frequently. Keep the rubbish area clean.
- Keep insects and rodents away from the kitchen – seal all holes and repair cracks.
- Have a first aid kit.



Faecal-Oral Transmission Route



• The F-Diagram

- How good hygienic practices prevents disease transmission.
- ☐ Primary barrier – prevents contact with faeces eg. good sanitation facilities (toilets)
- ☐ Secondary barrier – prevents faeces from being ingested eg. hand hygiene, Safe drinking water, food hygiene

Assignment

Pollution is caused mainly through anthropogenic activities

- a) List the contribution you and your community play in making drinking water unwholesome
- b) What are the causes of poor solid waste management in your community
- c) Ghana should ban plastics like the other countries eg Rwanda and Tanzania. In your opinion what impacts will it have on your life style as a student and as community
- d) How is waste water managed in your community? Describe it.
- e) What environmental effects do you foresee on the air quality and water bodies?